SILTHAUL · REGULATORY REGISTER

SiltHaul regulatory and standards register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/silthaul		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-03	Amish Chadha	First register, written for the kit 1.9.1 upgrade; standards checked on their issuing pages on 2026-11-03

SiltHaul regulatory and standards register

SiltHaul is a hand-powered rope and chain machine for flood clean-up; no product regulation was found that certifies it as a whole, and the design leans on lifting-accessory and machinery-safety standards for its rope, sling, shackle and guard choices. This is not legal advice; anything marked "to assess" needs a qualified reviewer before any product claim. Method: STANDARDS section 23.

Table 1. Regulatory and standards register.

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
G1	ISO 12100:2010 Safety of machinery, general principles for design, risk assessment and risk reduction (confirmed 2022)	International	Method for hazard identification, risk estimation and risk reduction in machine design	The preliminary hazard analysis and design FMEA in SLH-RSK-001 follow its order: remove by design, guard, then inform	Design lead	Applies; used as method (SLH-RSK-001)
G2	ISO 14120:2015 Safety of machinery, guards, general requirements for the design and construction of fixed and movable guards	International	Fixed and movable guards against mechanical hazards	The closed chain guard (BOM 9, PR-016); the ratchet wheels and pawls stay on the outer face of the frame	Design lead	Met by design for the chain guard; guard openings and fixings to assess at TRL 4
G3	ISO 11228-2:2007 Ergonomics, manual handling, Part 2: pushing and pulling (confirmed 2021, amendment 2022)	International	Recommended limits for whole-body pushing and pulling	Crank force (PR-002), tipping bar (PR-017) and, for option B, the 189 N each hauling pull (SLH-STM-001 S19)	Design lead	To assess: tables not read; check at TRL 4
G4	EN 1492-2:2000+A1:2008 Textile slings, safety, roundslings made of man-made fibres	EU, widely used elsewhere	Rating and testing of polyester roundslings up to 40 t; excludes lifting of people	The 2 t round sling that anchors the capstan to a tree or vehicle (BOM 12); used here as a pulling anchor, not for lifting	Design lead	Applies to the bought sling; use as an anchor to assess
G5	EN 13889:2003+A1:2008 Forged steel shackles for general lifting purposes, dee shackles and bow shackles, grade 6	EU, widely used elsewhere	Dimensions and working load limits of grade 6 shackles with threaded pins, 0.5 t and above	Bow shackles on the sling and the box bridles (BOM 12, 20); the bought shackles have 19 mm pins and cotter or split pins in the model	Design lead	To assess: the model's split-pin shackles are not threaded-pin grade 6 shackles; buy rated shackles at TRL 4
G6	ISO 606:2015 Short-pitch transmission precision roller and bush chains, attachments and sprockets (confirmed 2021)	International	Dimensions, strength and sprockets of 08B-1 chain	The 4:1 chain drive (BOM 6, 7); breaking load 17.8 kN from the maker's table	Design lead	Applies; met by design (SLH-CAL-001 J2)
G7	ISO 4017, ISO 4032 and ISO 7089 hex bolts, nuts	International	Dimensions of the bolts, nuts and	The 161 fasteners in the 245-part model of option A; numbers as	Design lead	

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
	and washers, as named by the CAD parts library		washers in the model	named in `.kit/cad_parts.py`, not checked against the standards this session		Applies; to check at TRL 4
G8	Flood clean-up health advice, for example Better Health Channel, State of Victoria (reviewed 2024-05-20) and OSHA flood response	Victoria, Australia; United States	Boots, gloves, hand washing, tetanus and contamination; manual lifting injuries	The safety notes and the briefing card (BOM 22); user behaviour, not a product requirement	Project lead	Applies as guidance; local health advice governs
G9	Machinery or equipment safety regulation of the country of use	To assess	Whether a hand-powered kit lent to volunteers is "machinery" in law	None identified; no CE-style marking is claimed	Regulatory consultant	To assess
G10	Rules for anchoring into floors of damaged or rented buildings, and for working in a flooded building	Local	Permission to drill the slab, structural safety, electricity	The tail block needs four drilled anchors in a sound slab (PR-007); the capstan stands on the ground	Co-design partner	To assess locally
G11	Disposal of flood mud and wastewater	Local	Where mud may be dumped	Out of scope of the design (SLH-PRB-001); option C would add 2.8 times the volume	Co-design partner	Does not apply to the design; to assess locally
G12	CERN-OHL-S-2.0 and MIT licences; REUSE	International	Licensing of the hardware and calculators	`REUSE.toml`, `LICENSES/`	Project lead	Met by design

Status: to assess, applies, does not apply, met by design, certification needed later.

Sources

- ISO, ISO 11228-2:2007, https://www.iso.org/standard/26521.html, checked 2026-11-03.
- ISO, ISO 12100:2010, https://www.iso.org/standard/51528.html, checked 2026-11-03.
- ISO, ISO 606:2015, https://www.iso.org/standard/61232.html, checked 2026-11-03.
- SIS (Swedish Standards Institute), ISO 14120:2015, https://www.sis.se/en/produkter/environment-health-protection-safety/safety-of-machinery/iso141202015, checked 2026-11-03.
- Genorma, EN 1492-2:2000+A1:2008, https://genorma.com/en/standards/en-1492-2-2000-a1-2008/amp, checked 2026-11-03.
- SIS, SS-EN 13889+A1:2008, https://online.sis.se/en/produkter/maskinsakerhet-357f77b1/chains-ropes-webbing-slings-and-accessories/ssen13889a12008, checked 2026-11-03.

- Better Health Channel, "After a flood: returning home safely", https://www.betterhealth.vic.gov.au/health/healthyliving/after-a-flood-returning-home-safely, reviewed 2024-05-20, checked 2026-11-03.
- OSHA, flood response, https://www.osha.gov/flood/response (cited in SLH-PRB-001).